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EXECUTIVE SUMMARY

Predictive and Autonomous Multi-Use Defense Architecture

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UNPROVEN AUTONOMY IS LIABILITY

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Current drones fly, but they don't think. Commander Qluu closes the gap between automated flight and true cognitive warfare. We don't just deliver hardware; we deliver a pilot that has flown 10 million sorties before deployment.

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ASYMMETRIC TOOLS FOR AN ASYMMETRIC FIGHT

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Commander QLUU

Cognitive Lethality: Powered by a proprietary V-JEPA world model that predicts threat movement 2–5 seconds into the future. Achieves Level 4 autonomous engagement to neutralize drone swarms with 5-meter terminal precision.

War Planner Model

Predictive Defense: A closed-loop ELO-adaptive engine that wargames tactical doctrine in latent space. Generates ranked Courses of Action (COAs) and continually learns via After Action Reports.

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COMMANDER QLUU MODEL

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Commander's Eye (V-JEPA)

Adapted from Meta's architecture, our Graph-JEPA model doesn't just react; it imagines. The UI visualizes the split between **Reality** and **Imagination**, predicting swarm movements 2-5 seconds ahead with 94.2% spatial accuracy.

Bayesian Threat Tracker

A custom 100-particle filter utilizing weather-adjusted sensor noise. Classifies enemy control modes (Human vs AI vs Pre-programmed) via heading variance with **1.2-second detection-to-track** latency.

Swarm Coordinator

Multi-drone collaborative intercept planner over a 500m mesh network. Utilizes Kruskal's MST routing to achieve coordinated, multi-vector target convergence and execute the **Cannonball Protocol**.

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HOW IT WORKS: COMMANDER QLUU

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1 OSINT Integration

Autonomous feed harvesting converts raw intelligence into 128-dimensional context vectors injected directly into the JEPA encoder, grounding the world model.

2 Bayesian Tracking

A custom 100-particle filter tracks targets under severe weather degradation, achieving an unprecedented **1.2-second detection-to-track** latency.

3 Data Layer

Our Graph-JEPA architecture processes swarm data in latent space. Instead of reacting, it accurately predicts threat movements **2–5 seconds ahead.**

4 Orchestration

Achieves **320ms decision latency**. Executes the Cannonball Protocol for multi-vector intercepts, yielding 5-meter impact precision and a 92% success rate.

Additionally:

Executes defensive playbooks through parallel agentic workflows and via proprietary predictive models.

Operates reliably across fragmented, noisy, and incomplete data environments through autonomous entity resolution.

Serves as the connective tissue between disparate data nodes and persistent, evolving swarm profiles.

Enables end-to-end tactical defense: search → match → predict → act.

Purpose-built configurations for military installations, intelligence, and critical infrastructure security.

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WAR PLANNER MODEL

AI TRAINING DASHBOARD

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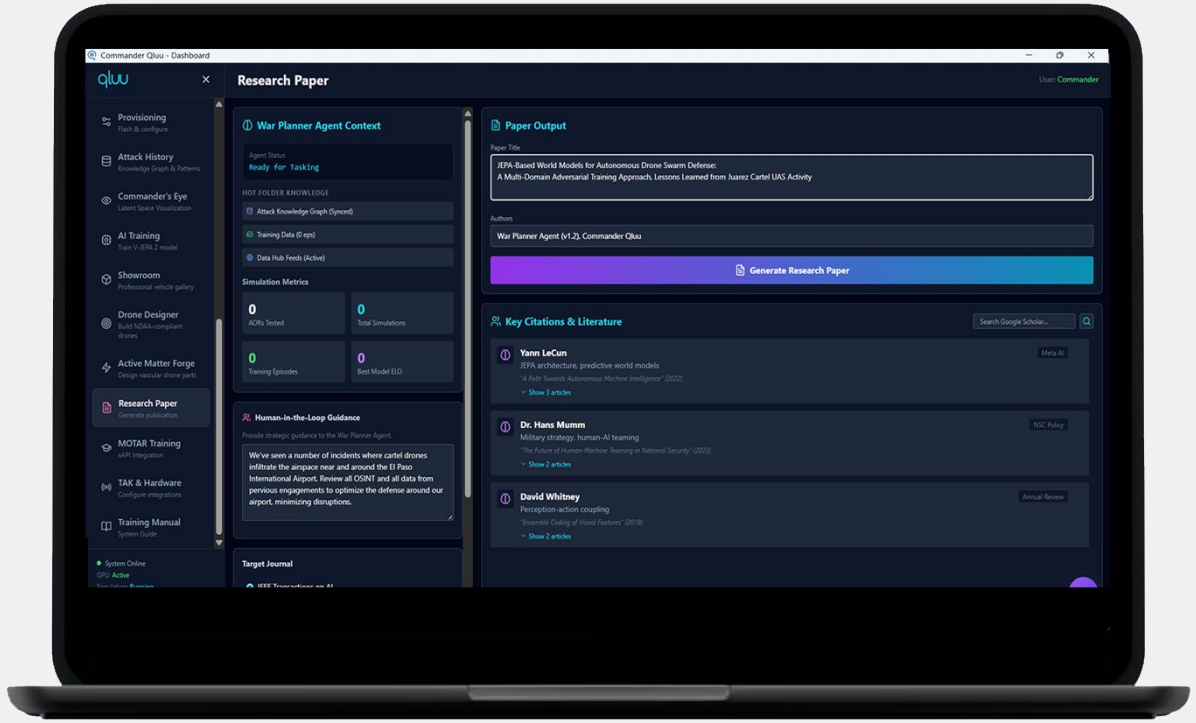
Strategy Generation & Self-Improvement

- **LLM-Based Strategist:** Generates ranked COAs with calculated risk levels (pWin, estimated losses). Synthesizes simulation data into formal Research Papers and tactical doctrine instantly.
- **Drone Designer & CFD:** Build entirely new, NDAA-compliant UAS assets within QLUUos. Test aerodynamics directly via the integrated OpenFOAM wind-tunnel simulation before 3D printing.
- **Hardware Agnostic (No Lock-in):** Unlike $28B competitors requiring proprietary Sentry towers, QLUUos is a 109,000+ file stack (React/Fastify/PostGIS) that deploys to existing drone fleets (MAVLink/Skydio) in hours.



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HOW IT WORKS: WAR PLANNER

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3D World Simulation

Ingests photorealistic terrain tiles. Tests physical hardware deployments against simulated swarms and active EW (Electronic Warfare) jammers to ensure operational readiness before live execution.

ELO Adaptive Difficulty

Our closed-loop self-improvement engine. A 3-tier rating system where Adversary AI organically escalates tactics (pincer movements, swarms) as Commander Qluu improves, forging "Grandmaster" readiness.

MOTAR Training Integration

Natively syncs with Air Force MOTAR VR training platforms. Operators track readiness globally via proficiency matrices and leaderboards, seamlessly bridging AI tactics with human oversight.

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HOW IT WORKS: WAR PLANNER

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10,000 Hours of Combat in 10 Hours of Compute.

Our **`Simulation Engine`** validates tactics without risking a single airframe.

Adaptive Adversary: The AI trains against an enemy that cheats, swarms, and creates electronic warfare (EW) environments.

Physics-Based Constraints: We simulate battery drain, wind shear, and sensor degradation. If it works here, it works downrange.

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QLUU VALIDATION REPORT

COMMANDER QLUU PROTECTING EL PASO AIRSPACE

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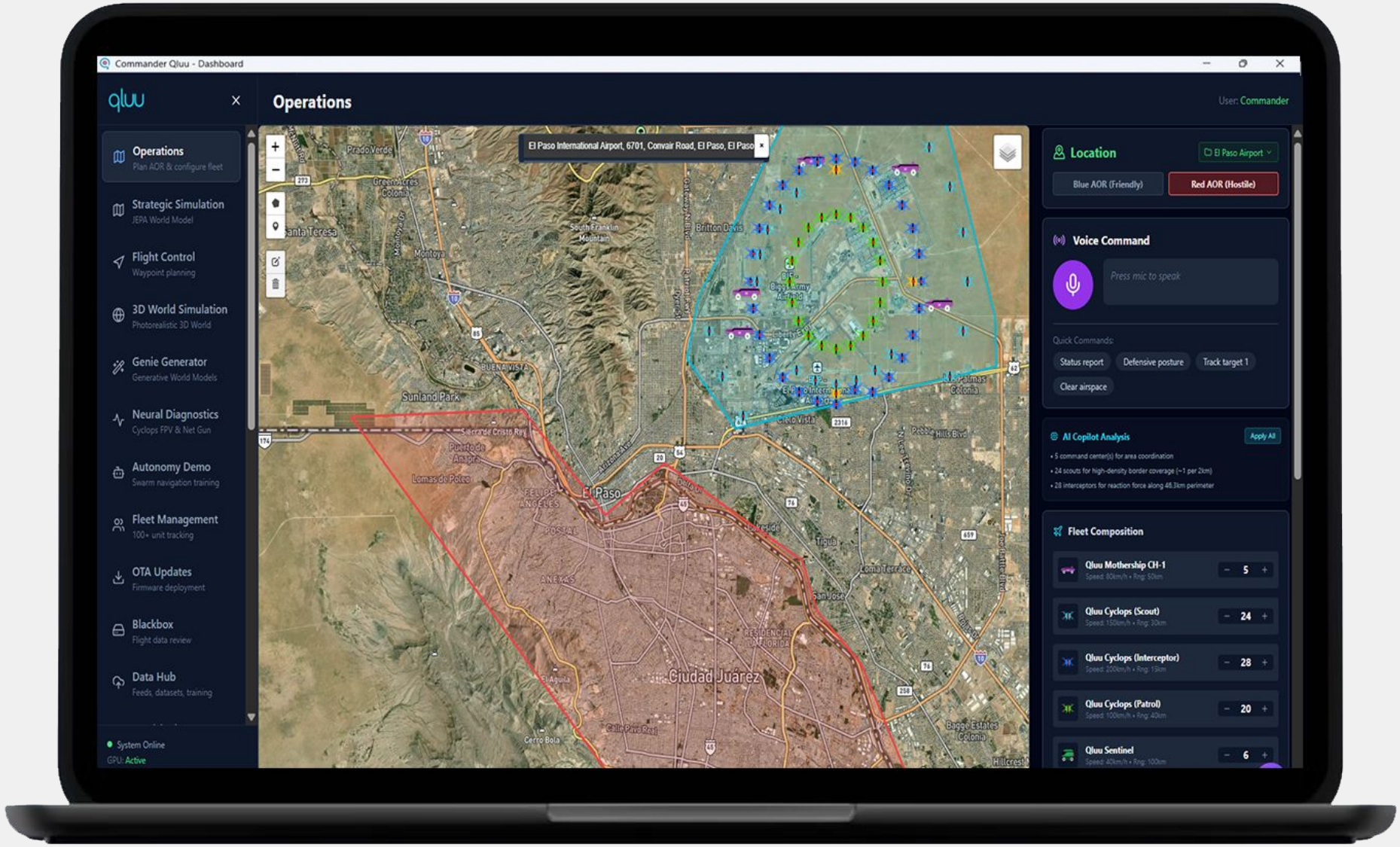

A Case Study in Autonomous Defense

KEY FINDINGS

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Key Findings	
Metric	**Count**
Total devices in both countries	11,192
Devices using unauthorized crossings	3,248
Signals in border zone	2.27 million

Top Unauthorized Crossing Hotspots			
Location	**Coordinates**	**Devices**	**Notes**
Juarez area (Mexico)	31.738, -106.485	637	Largest hotspot - NE of central crossing
El Paso/Loop 375 (US)	31.762, -106.421	490	Between highway and border wall
Juarez area 2	31.730, -106.480	320	Adjacent to hotspot 1
Sunland Park area	31.796, -106.556	237	Near Rio Grande river
Socorro (Texas)	31.637, -106.271	127	Southern crossing point

TRACKING THE CARTEL SWARM

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Analyzing the Incursion Pattern in El Paso

- **Mass Surveillance:** Cartels deployed daily autonomous swarms to monitor Border Patrol shifts. The density directly beneath commercial descent paths into El Paso Int'l created a severe aviation hazard.
- **RF Silence Defeats Legacy Systems:** Swarms operated entirely on Visual-Inertial Odometry (VIO). With no active RF links, traditional multi-million dollar jammers were completely neutralized.
- **Weather-Resilient Tracking:** Despite dynamic weather dropping visibility to 60% in storm conditions, Qluu's Bayesian Particle Filter maintained **87% track confidence** on the swarms.

DETECT-TO-TRACK **1.2 SEC**

CONTROL CLASSIFICATION **AUTONOMOUS**

THE STRATEGIC STANDOFF

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Pentagon Proposes Directed Energy With RF jamming rendered useless, the military proposed High-Power Microwave (HPM) and lasers to burn the drones out of the sky.

FAA Objects: Airspace Closure The FAA declared military lasers a catastrophic blinding risk to commercial airline pilots. To allow the military to operate safely, the FAA ordered the total grounding of commercial flights in El Paso.

The Hardware Lock-in Failure The cartel inadvertently paralyzed U.S. infrastructure. Grounding flights to allow rigid, hardware-centric defense systems a clear shot proved that legacy approaches are too slow, costly, and disproportionate for modern asymmetric threats.

THE QLUU RESOLUTION

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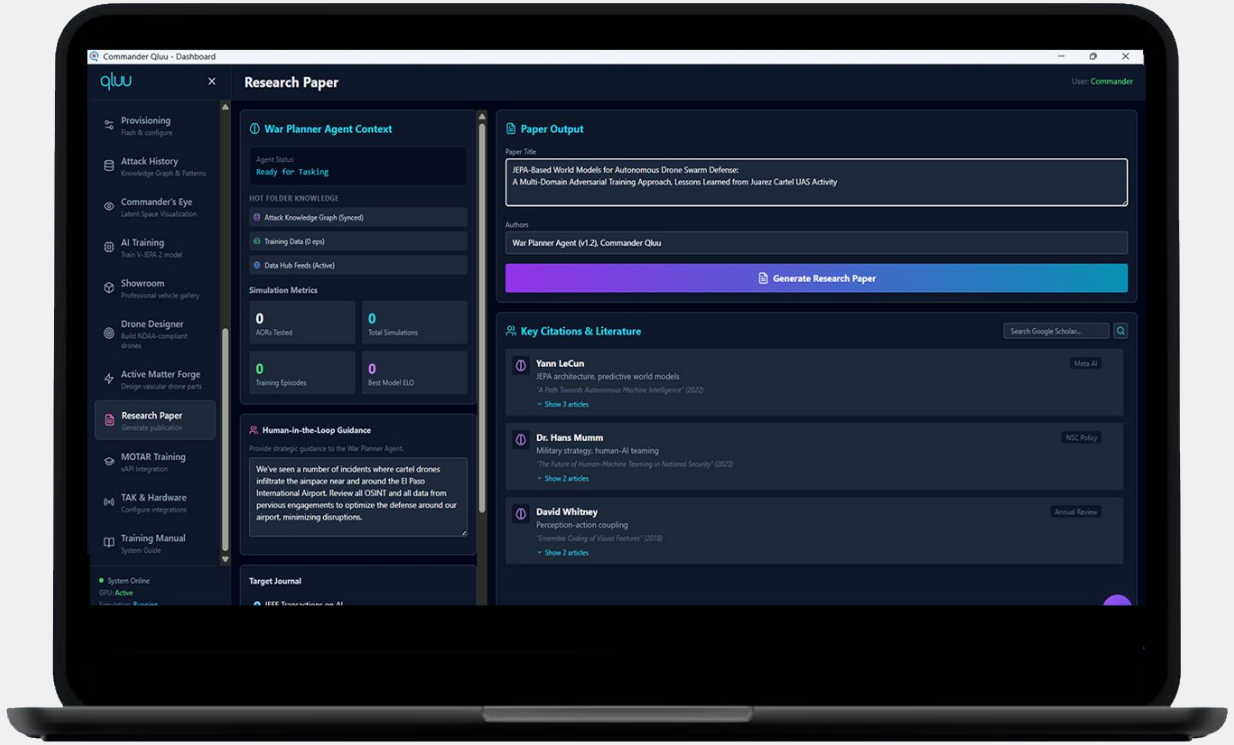
Surgical Intercept Sequence

- **Rapid Simulation:** QLUUos ingested El Paso DSM terrain and FAA flight corridors, running thousands of synthetic episodes to calculate safe physical intercept vectors outside of pilot sightlines.
- **Hardware-Agnostic Launch:** Operating over a 500m mesh network, existing Border Patrol Cyclops UAVs were integrated into QLUUos instantly—no new towers built.
- **Kinetic Takedown:** Executing the **Cannonball Protocol**, the swarm coordinator achieved 5-meter terminal impact precision, physically neutralizing the cartel drones.
- **Airspace Reopened:** Zero lasers fired. The FAA immediately lifted grounding orders.



STRATEGIC VALIDATION

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Let Us Prove the ELO in Your Theater

Unlike hardware-centric primes with 12-month installation cycles, QLUUos is a containerized stack that deploys in hours. Let us map your infrastructure.

We are ready to validate the Commander Qluu Cognitive Agent against your specific mission sets.

Phase 1: Ingest your specific AOR terrain data.

Phase 2: Train the Agent to "Grandmaster" ELO in simulation.

Phase 3: Live fly-off demonstration demonstrating the trained behaviors.

STRATEGIC VALIDATION

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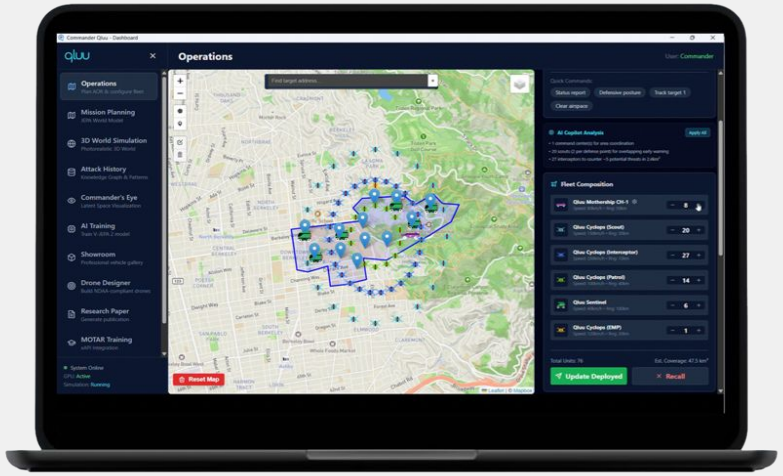
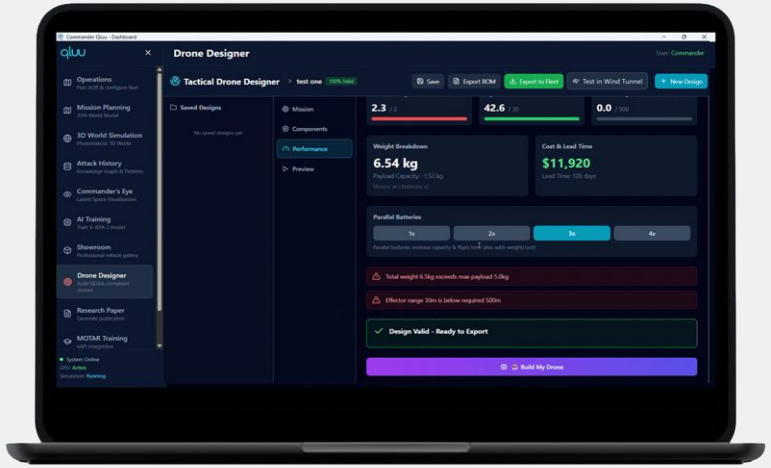


ELO as the Standard of Lethality



AOR-Specific Training (The "Home Court" Advantage)

The Simulation Forge (Sim-to-Real)

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Protecting What Matters, Autonomously.

THANK YOU

Arkadiy Okhman
Founder – CEO
ao@qluulab.com

Mike Jacobs
Co-Founder – CTO
mike@qluulab.com